A Project Report on

MEDICAL IMAGE SEGMENTATION FOR BRAIN LESION DELINEATION AND QUANTITATIVE VOLUMETRICS USING DEEP LEARNING (U-NET & FCN-8S)

For **Mini Project (AI&ML in Healthcare: Lab)** of Final Year, **(BE Sem - VII)**

Course code: **HAIMLSBL701** in
Electronics and Telecommunication Engineering

by

Suraj Ajay Chauhan (XC-23-0314)

Under the guidance of

Dr. Vijay Galshetwar



Hope Foundation's

Finolex Academy of Management and Technology, Ratnagiri

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CERTIFICATE

This is to certify that the project work entitled **"Medical Image Segmentation for Brain Lesion Delineation and Quantitative Volumetrics Using Deep Learning (U-Net & FCN-8s)"** is a bonafide work of

Suraj Ajay Chauhan (Registration No.: XC-23-0314)

submitted to the University of Mumbai in partial fulfillment of the requirements for **Mini Project (Course Code: HAIMLSBL701 — AI&ML in Healthcare: Lab)** of Final Year, **(BE Sem - VII)** in **Electronics and Telecommunication Engineering** as laid down by the University of Mumbai, during academic year **2026–26**.

Dr. Vijay Galshetwar (Project Guide)	**Dr. S. V. Chougule** (H.O.D. EXTC)

Date: ...../...../2026 | **Place:** Department of Electronics & Telecommunication Engineering, FAMT, Ratnagiri.

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Suraj Ajay Chauhan (XC-23-0314)
(BE Sem-VII, Department of EXTC, FAMT, Ratnagiri)

ABSTRACT

Manual delineation of lower-grade gliomas (LGG) on magnetic resonance imaging (MRI) is slow and varies between observers, yet accurate lesion boundaries and measurements are needed for surgical planning, radiotherapy targeting and follow-up. This project develops a deep-learning pipeline for pixel-level segmentation of FLAIR-abnormal brain lesions together with an interactive software workstation (the MedSeg Clinical Workstation). Two architectures were implemented and benchmarked: a U-Net with skip connections (about 7.76 million parameters) trained from scratch, and an FCN-8s with a pretrained VGG16-BN backbone (about 20.49 million parameters). Both were trained with a hybrid binary cross-entropy and soft Dice loss on the TCGA-LGG dataset of 110 patients and 3,929 axial slices (T1 pre-contrast, FLAIR, T1 post-contrast), using a strict patient-level split (78/16/16 patients) to prevent data leakage.

On 540 unseen test slices, FCN-8s achieved an all-slice Dice of 0.8749 (tumour-slice Dice 0.6585) and a slice specificity of 99.43%, while the U-Net achieved 0.8014 (0.5976), a tumour precision of 85.25%, a specificity of 91.38% and a median 95th-percentile Hausdorff distance of 8.99 pixels. A run without augmentation produced false alarms on every tumour-free slice, which shows that specificity must be reported alongside Dice; because that run was a smaller network, the result is indicative. A single-channel greyscale model extends inference to scans from other sources, at lower specificity. The workstation supports DICOM, NIfTI, TIFF and PNG/JPG input, window/level control, area, effective-diameter and estimated-volume measurements, multi-model comparison and one-click A4 PDF reports. It is a research prototype and has not been clinically validated.

Keywords: medical image segmentation; U-Net; FCN-8s; TCGA-LGG; brain MRI; deep learning; Dice similarity coefficient; Hausdorff distance; quantitative volumetrics; PACS-style workstation.

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LIST OF ABBREVIATIONS

Abbreviation	Expansion	Abbreviation	Expansion
AMP	Automatic Mixed Precision	**GTV**	Gross Tumour Volume
BCE	Binary Cross-Entropy	**HD95**	95th-percentile Hausdorff Distance
BN	Batch Normalisation	**IMRT**	Intensity-Modulated Radiotherapy
CAD	Computer-Aided Diagnosis	**IoU**	Intersection over Union
CNN	Convolutional Neural Network	**LGG**	Lower-Grade Glioma
CSF	Cerebrospinal Fluid	**MRI**	Magnetic Resonance Imaging
CTV	Clinical Target Volume	**NIfTI**	Neuroimaging Informatics Technology Initiative
CUDA	Compute Unified Device Architecture	**PACS**	Picture Archiving and Communication System
DICOM	Digital Imaging and Communications in Medicine	**RANO**	Response Assessment in Neuro-Oncology
DSC	Dice Similarity Coefficient	**ReLU**	Rectified Linear Unit
FCN	Fully Convolutional Network	**SRS**	Stereotactic Radiosurgery
FLAIR	Fluid-Attenuated Inversion Recovery	**TCGA / TCIA**	The Cancer Genome Atlas / Imaging Archive
FP16 / GPU	16-bit Floating Point / Graphics Processing Unit	**TTA / WHO**	Test-Time Augmentation / World Health Organization

CHAPTER 1 INTRODUCTION

1.1 Need for the Selected Topic

Diffuse lower-grade gliomas (LGG), i.e. WHO grade 2 and 3 astrocytomas and oligodendrogliomas [1], are among the most challenging tumours in clinical neuro-oncology. They arise from the glial supporting tissue of the brain and grow by infiltrating along white-matter tracts, so their margins blend imperceptibly into healthy brain parenchyma. Large-scale genomic profiling has shown that LGGs fall into molecularly distinct subtypes [2], and the shape of the tumour on MRI has been associated with these subtypes [3]. Accurate and reproducible delineation is therefore valuable for research as well as for patient care.

Magnetic Resonance Imaging (MRI) is the established non-invasive modality for diagnosing and monitoring brain tumours. Routine protocols acquire several complementary sequences:

- **T1-weighted (native):** anatomical detail and grey/white-matter contrast.
- **T1-weighted contrast-enhanced (T1-Gd):** gadolinium-based contrast highlights breakdown of the blood–brain barrier and so outlines actively enhancing tumour.
- **T2-weighted:** fluid collections and oedema appear bright.
- **Fluid-Attenuated Inversion Recovery (FLAIR):** suppresses the free cerebrospinal-fluid (CSF) signal while preserving T2 prolongation, which makes peritumoral oedema and infiltrative tumour clearly visible.

The dataset used in this project provides three of these sequences (T1 pre-contrast, FLAIR and T1 post-contrast) together with manually drawn FLAIR-abnormality masks (Section 3.4).

Despite advances in image acquisition, lesion delineation in routine practice is still largely manual or semi-manual. The radiologist or radiation oncologist traces the boundary slice by slice through a volume of dozens to hundreds of images. This paradigm has four main drawbacks:

1. **Time and fatigue.** Contouring a complex lesion through a whole volume is slow, often tens of minutes per patient, and becomes a bottleneck in busy centres.
2. **Observer variability.** Even experts disagree. In the BRATS benchmark, the Dice agreement between human raters on glioma sub-regions was only about 74–85% [4].
3. **Need for accurate margins.** In surgery the aim is maximal safe resection: removing too little leaves infiltrating tumour behind, whereas removing too much risks damage to eloquent cortex. In stereotactic radiosurgery (SRS) and intensity-modulated radiotherapy (IMRT), the gross tumour volume (GTV) and clinical target volume (CTV) must be defined accurately so that the dose reaches the tumour while sparing the brain stem, optic chiasm and hippocampus.
4. **Need for standardised volumetrics.** Response criteria such as RANO rely on bidimensional measurements, i.e. products of perpendicular diameters [5], [6]. Volume measurements can capture changes that diameter products miss, but they are impractical to obtain without automatic segmentation.

A robust, automated computer-aided diagnosis (CAD) system that segments brain lesions with high spatial fidelity, reports quantitative measurements and fits into a radiology-style workflow is therefore a worthwhile engineering goal. This report describes the design, training, evaluation and software implementation of such a system, built around two deep-learning architectures: U-Net and FCN-8s.

1.2 Description

This project develops an end-to-end deep-learning framework for automated brain-lesion segmentation and quantitative analysis, and wraps it in an interactive software workstation. The emphasis is on moving from a trained model to a usable tool: the models are trained and evaluated under a leakage-free protocol, and the results are exposed through a PACS-style interface that allows the outputs to be inspected, compared and exported.

1.2.1 Core Theoretical Principles

Image segmentation is formulated as dense, pixel-level binary classification. Let $\mathbf{X} \in \mathbb{R}^{H \times W \times C}$ be an input slice, where H = W = 256 and C = 3 for multi-sequence input (T1 pre-contrast, FLAIR, T1 post-contrast) or C = 1 for single-channel greyscale scans. The objective is to learn a parameterised mapping:

$$f_\theta : X \mapsto \hat{Y} \in [0, 1]^{H \times W} \tag{1.1}$$

where $\hat{Y}_{i,j} = P(Y_{i,j} = 1 \mid X)$ is the posterior probability that pixel (i, j) belongs to the lesion class and θ denotes the trainable weights. A binary mask is obtained by thresholding the probability map at a decision threshold τ. Two architectural paradigms are investigated:

- **U-Net [7]:** an encoder–decoder network with a symmetric contracting and expanding path, joined by long skip connections that concatenate high-resolution encoder feature maps with the corresponding decoder stages. The skip connections bypass the information bottleneck and preserve the fine spatial detail needed for sharp boundaries.
- **Fully Convolutional Network (FCN-8s) [8]:** a classification backbone (here VGG16 with batch normalisation [9], [10]) converted into a fully convolutional network. FCN-8s adds multi-scale predictions by fusing score maps taken from the pool3, pool4 and conv7 stages.

1.2.2 End-to-End System Capabilities

The software is not only a model-evaluation script; it is a complete workstation prototype (the MedSeg Clinical Workstation, implemented in Streamlit) with multi-format data ingestion (DICOM .dcm including rescale slope/intercept and header metadata, NIfTI .nii/.nii.gz, TIFF, PNG and JPG up to 200 MB), PACS-style viewing controls (window width/level, probability-threshold slider, optional horizontal-flip TTA, and contour/filled/heatmap overlays), quantitative measurements (pixel count, cross-sectional area in mm² and cm², equivalent circular diameter, single-slice spherical volume estimate in cm³, and centroid X, Y), project-defined size categorisation, a Multi-Model Comparator tab, a browsable 110-patient TCGA-LGG Clinical Database tab, automated one-page A4 PDF

CHAPTER 2 LITERATURE SURVEY

2.1 Survey of Existing Systems

Automated medical-image segmentation has passed through four broad algorithmic eras: intensity-based methods, deformable models, classical machine learning with hand-crafted features, and deep convolutional networks (Figure 2.1).

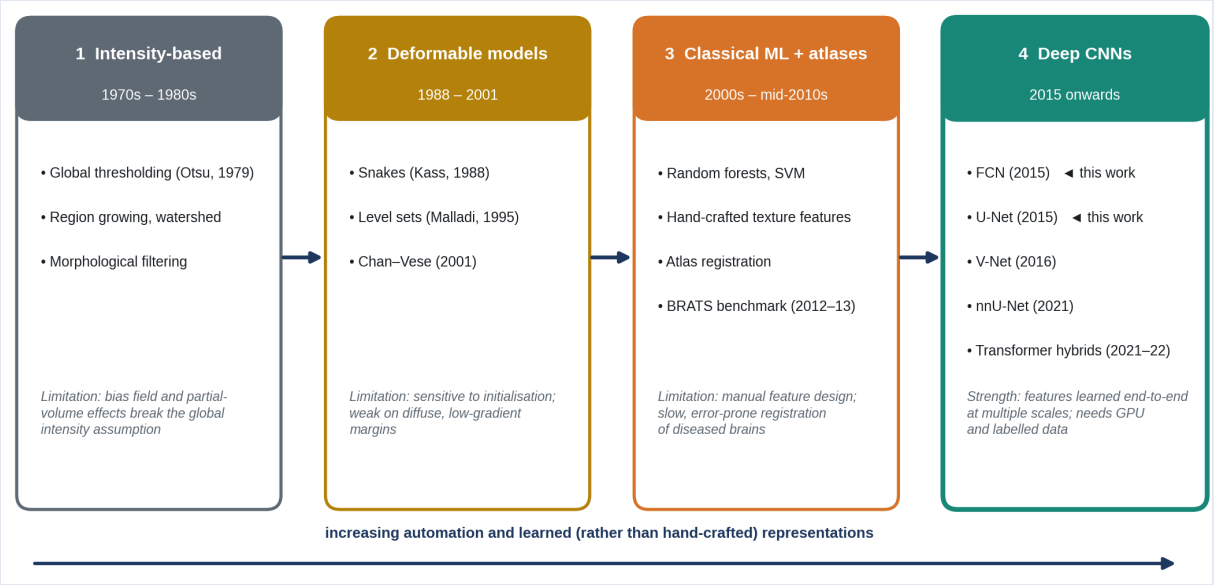


Figure 2.1: Evolution of medical image segmentation, from intensity-based methods to deep convolutional networks.

1. Intensity Thresholding and Morphological Filtering. The earliest approaches relied on histogram thresholding, for example Otsu's automatic global threshold [11], together with region growing, watershed transforms and morphological filtering (erosion, dilation). They assume that pathological tissue forms an intensity population clearly separable from healthy grey and white matter. *Limitations:* In brain MRI, radio-frequency coil inhomogeneity produces a slowly varying bias field, and partial-volume averaging mixes CSF, white matter and oedema within a single voxel, producing fragmented contours and false positives.

2. Deformable Models and Active Contours. Active contours ("snakes") [12] and level-set methods [13], [14] cast segmentation as energy minimisation: an internal term keeps the curve smooth, while an external term attracts it to image edges. *Limitations:* Sensitive to initialisation and weak on infiltrative, non-enhancing glioma margins where no clear intensity gradient exists.

3. Classical Machine Learning and Atlas Registration. Supervised classifiers (random forests, SVMs) on hand-crafted voxel features (Gabor filters, local binary patterns, texture statistics) combined with registered brain atlases. In the BRATS benchmark [4], no single algorithm ranked first for every sub-region. *Limitations:* Hand-crafted features miss multi-scale context, and deformable registration of mass-effect-distorted brains is slow and error-prone.

4. Deep Learning and Fully Convolutional Networks. Long et al. [8] turned classification networks into dense predictors (FCN-8s); Ronneberger et al. [7] introduced U-Net with long skip connections for precise pixel localisation; V-Net [15] introduced the Dice loss; nnU-Net [16] demonstrated strong self-configuring pipelines, and transformer hybrids (TransUNet [17], Swin UNETR [18]) add global context.

Table 2.1: Comparative synthesis of existing medical image segmentation techniques.

Methodology	Representative Algorithms	Advantages	Critical Drawbacks	Refs.
Thresholding & morphology	Otsu thresholding, region growing, watershed	Computationally light; simple to implement	Vulnerable to bias field, noise and partial-volume effects	[11]
Deformable	Snakes, level sets, Chan–	Enforces smooth, closed	Sensitive to initialisation;	[12]–

2.2 Problem Statement and Objectives

2.2.1 Problem Statement

Given an axial brain MRI slice $\mathbf{X}$ acquired under diverse protocols – either a multi-contrast composite (T1 pre-contrast, FLAIR, T1 post-contrast) or a single-channel greyscale image – design, train and validate an automated deep convolutional system that segments the lesion boundary $\hat{\mathbf{Y}}$ and reports quantitative measurements through an interactive workstation.

> **Quantitative Targets of the Project:**
>
> - **All-slice mean Dice similarity coefficient ≥ 0.80** on unseen test patients (defined in Section 4.1; slices that are correctly predicted empty score 1);
> - **Median 95th-percentile Hausdorff distance (HD95) ≤ 10 pixels**;
> - **Slice-level specificity ≥ 90%**, i.e. false alarms on at most 10% of tumour-free slices.

2.2.2 Key Engineering Challenges

1. **Extreme class imbalance.** In a 256 × 256 slice (65,536 pixels) a lesion typically occupies a few hundred to a few thousand pixels (roughly 200–3,000 pixels, i.e. 0.3–4.6% of the slice), and a large share of slices contain no lesion at all (348 of the 540 test slices, 64%). Unweighted pixel-wise losses such as mean squared error or plain binary cross-entropy are dominated by background and can converge to the trivial all-background solution.
2. **Data leakage in volumetric data.** Adjacent slices from the same patient are nearly redundant. If slices of one patient appear in both the training and the test set, the network can memorise patient-specific anatomy and report inflated metrics that collapse on new patients. A patient-level split is therefore essential.
3. **Cross-modality generalisation.** The multi-sequence models are trained on three-channel composites (red = T1 pre-contrast, green = FLAIR, blue = T1 post-contrast). External scans are usually single-channel greyscale (R = G = B), so any inter-channel contrast the model has learned to exploit is absent. A dedicated single-channel model is needed (Section 4.4).

2.2.3 Project Objectives

1. Curate the TCGA-LGG subset (110 patients, 3,929 axial slices) and establish a reproducible, strict patient-level split (78 training / 16 validation / 16 test patients) with no cross-patient contamination.
2. Implement, train and optimise two architectures – a U-Net trained from scratch and an FCN-8s with a pretrained VGG16-BN backbone – using a hybrid BCE + soft-Dice loss, AdamW optimisation [19] and mixed-precision training [20].
3. Study the effect of online data augmentation (flips, affine transformations, brightness/contrast jitter) on false alarms in tumour-free slices by means of an ablation run.
4. Train an adaptive single-channel greyscale model (`unet_general`) so that scans from other sources can be processed.
5. Build an interactive, PACS-style workstation with window/level control, 2D area and single-slice volume estimates, multi-model comparison and automatic PDF report generation.

CHAPTER 3 OVERVIEW OF THE SYSTEM AND IMPLEMENTATION

3.1 Introduction & 3.2 Block Diagram

The proposed system, the **MedSeg Clinical Workstation**, is a modular, end-to-end medical-image computing pipeline that connects deep-learning research code to a radiology-style user interface. It consists of four subsystems: **(A) Data ingestion and standardisation** (decodes DICOM, NIfTI, TIFF and raster files, normalises intensities and extracts patient metadata); **(B) Deep-learning segmentation** (holds the U-Net and FCN-8s models and performs the forward pass, optional TTA and thresholding); **(C) Quantitative volumetric analysis** (derives pixel count, area, equivalent diameter, volume estimate and centroid); and **(D) Presentation and export** (Streamlit interface and vector PDF report generator).

Figure 3.1 shows the data flow through six processing blocks: **Block 1 (Ingestion)** applies DICOM rescale slope/intercept, clips and min–max-normalises intensities, resizes to 256 × 256 by bilinear interpolation and routes channels; **Block 2 (Inference engine)** runs CUDA FP16 mixed-precision forward pass, optional horizontal-flip TTA and sigmoid mapping; **Block 3 (Post-processing)** thresholds at τ, removes noise and extracts contours; **Block 4 (Biomarker engine)** computes spatial metrics; **Block 5 (Interactive UI)** renders overlays and comparator; and **Block 6 (Report generator)** compiles the A4 PDF report.

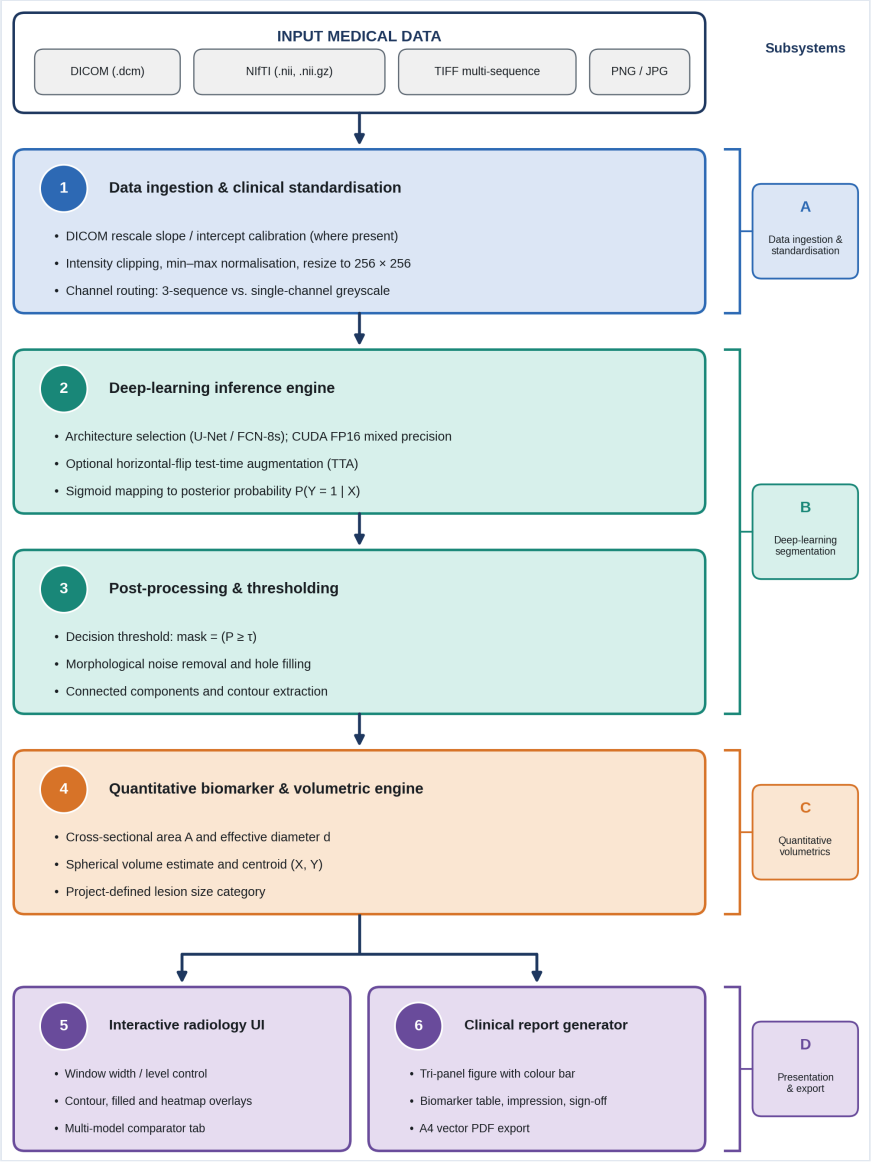


Figure 3.1: Block diagram of the MedSeg Clinical Workstation.

3.3 Computational Workflow and Working

3.3.1 Mathematical Loss Formulation

To counter the foreground–background imbalance, a compound loss combines binary cross-entropy (BCE) with soft Dice loss, the latter introduced for segmentation in V-Net [15], with $\lambda_{BCE} = \lambda_{Dice} = 0.5$ and smoothing constant $\varepsilon = 10^{-6}$:

$$L_{total}(y, p) = \lambda_{BCE}\, L_{BCE}(y, p) + \lambda_{Dice}\, L_{Dice}(y, p) \tag{3.1}$$

$$L_{BCE}(y, p) = -\,(1 / N) \sum_{i=1..N} [\, y_i \log(p_i) + (1 - y_i) \log(1 - p_i)\,] \tag{3.2}$$

$$L_{Dice}(y, p) = 1 - (2 \sum_{i=1..N} y_i\, p_i + \varepsilon) / (\sum_{i=1..N} y_i + \sum_{i=1..N} p_i + \varepsilon) \tag{3.3}$$

The BCE term evaluates each pixel independently and gives smooth gradients early in training; the soft Dice term measures overlap over the entire mini-batch at once. In the implementation (Listing 3.2) the BCE term operates on logits (`BCEWithLogitsLoss`) for numerical stability.

3.3.2 U-Net Architectural Topology

The implemented U-Net (Figure 3.2) has a contracting encoder (four blocks of two 3×3 convolutions, batch normalisation [10] and ReLU: $h_l = \mathrm{ReLU}(\mathrm{BN}(W_l * x + b_l))$ (Eq. 3.4), with 2×2 max-pooling halving spatial size as channels double $32 \to 64 \to 128 \to 256$), a bottleneck at $16 \times 16 \times 512$ with dropout ($p = 0.1$), and four decoder up-sampling blocks using 2×2 transposed convolutions $u_l = \mathrm{ConvT}(h_{l+1})$ (Eq. 3.5). Skip connections concatenate encoder and decoder maps along the channel axis $z_l = [h_{enc,l} \,\|\, u_{dec,l}]$ (Eq. 3.6, Figure 3.3) before a double convolution and a final 1×1 output convolution (≈ 7.76 million trainable parameters).

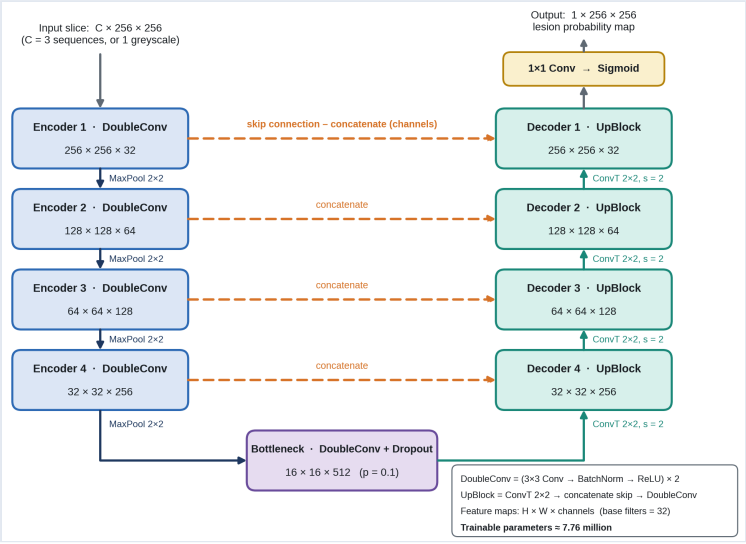


Figure 3.2: Architecture of the implemented U-Net (base width 32).

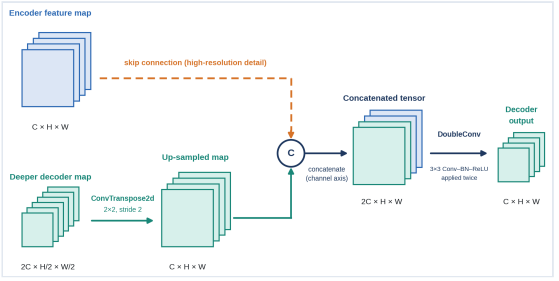


Figure 3.3: Skip-connection mechanism: encoder features concatenated with decoder features.

3.3.3 Fully Convolutional Network (FCN-8s) Topology

FCN-8s [8] (Figure 3.4) adapts a VGG16 network with batch normalisation [9], [10] into a fully convolutional segmenter initialised with pretrained torchvision VGG16-BN weights. Five VGG stages (Conv1–Conv5, 13 convolutional layers) end in 2 × 2 max-pooling with outputs of sizes 128^2, 64^2, 32^2 (`pool3`), 16^2 (`pool4`) and 8^2 (`pool5`). The classical fully connected layers are replaced by convolutional layers (`conv6`, `conv7`) with spatial dropout (p = 0.2). Multi-scale fusion projects `conv7` to a 1-channel score map via 1 × 1 conv, up-samples by 2, adds the 1 × 1-convolved `pool4` score map, up-samples by 2 again, adds the 1 × 1-convolved `pool3` score map, and up-samples by 8 (8 × 8 transposed conv, stride 8) to restore 256 × 256 resolution (≈ 20.49M parameters total; 14.72M in VGG16-BN).

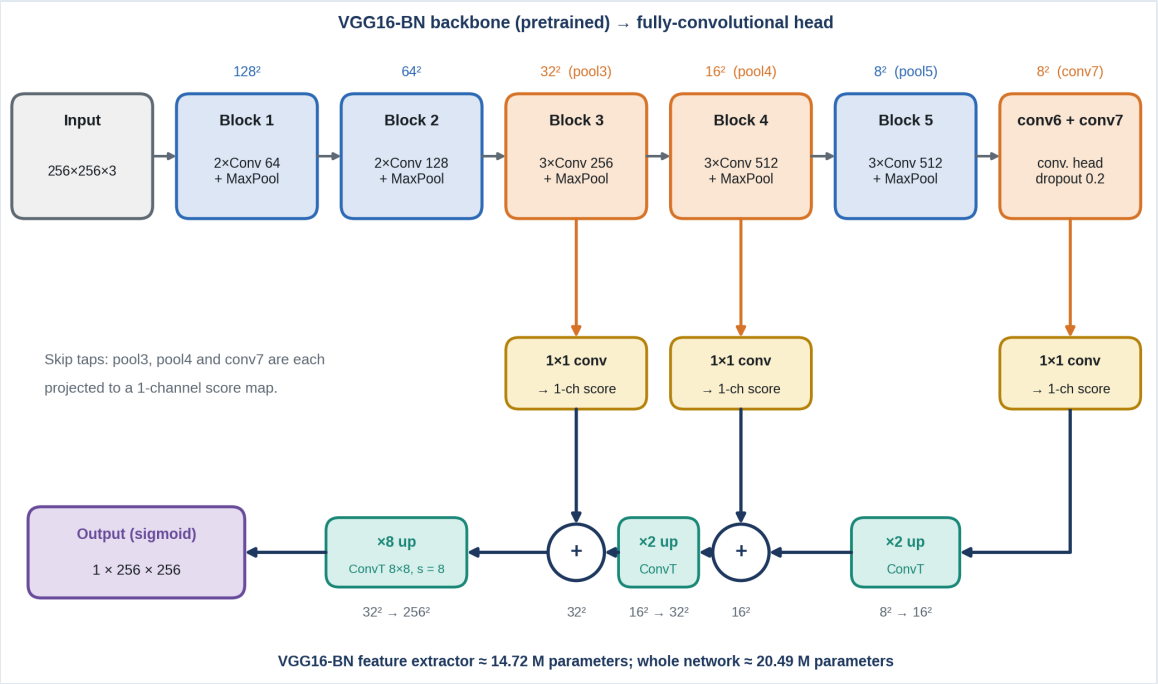


Figure 3.4: Architecture of FCN-8s with a VGG16-BN backbone and multi-scale skip fusion.

3.3.4 Quantitative Biomarker Mathematics

From the binary mask $M \in \{0, 1\}^{H \times W}$ (where $M_{i,j} = 1$ if predicted probability $\geq \tau$) and assumed pixel spacing $\Delta x = \Delta y = 1.0$ mm, the pixel count, cross-sectional area, effective circular diameter, spherical volume estimate and centroid coordinates are derived as:

$$N_{px} = \sum_{i=1..H} \sum_{j=1..W} M_{i,j} \quad | \quad A = N_{px} \cdot \Delta x \cdot \Delta y \ \ (\text{mm}^2, \text{divide by } 100 \text{ for cm}^2) \qquad \textbf{(3.7, 3.8)}$$

$$d_{eff} = 2\sqrt{(A / \pi)} \quad | \quad V_{est} = (4 / 3)\,\pi\, r_{eff}^3, \ \ r_{eff} = d_{eff} / 2 \ \ (\text{cm}^3) \qquad \textbf{(3.9, 3.10)}$$

$$X_c = (1 / N_{px}) \sum_{i=1..H} \sum_{j=1..W} j \cdot M_{i,j} \quad | \quad Y_c = (1 / N_{px}) \sum_{i=1..H} \sum_{j=1..W} i \cdot M_{i,j} \qquad \textbf{(3.11, 3.12)}$$

Important limitation: The volume estimate extrapolates a 3D volume from a single 2D slice by assuming a sphere. Real gliomas are irregular and span many slices, so V_{est} (e.g. 261.19 cm³ for a 49.40 cm² slice in Figure 4.5) is displayed only as an indicative size measure. Lesions are also assigned to one of four project-defined heuristic size labels (*Sub-centimetre Focal*, *Localized Lesion*, *Moderate Infiltration*, *Extensive Macro-Lesion*), which are software size categories and not WHO histological grades [1].

3.3.5 Training Pipeline and Data Partitioning

Figure 3.5 summarises the training procedure. All slices of a patient are assigned to exactly one subset (78 training / 16 validation / 16 test patients) to prevent slice-level leakage. Online augmentation is applied to training slices only, losses are optimised under FP16 automatic mixed precision [20] with AdamW [19], and `best.pt` is selected on validation Dice.

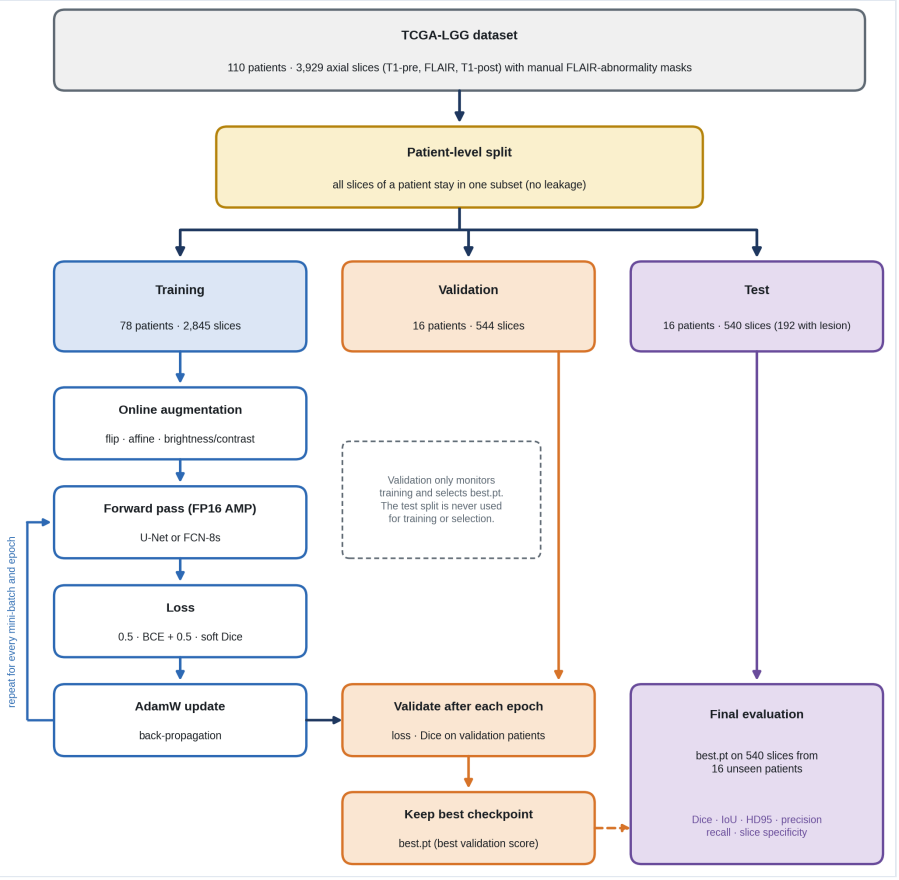


Figure 3.5: Data partitioning, online augmentation and mixed-precision training pipeline.

3.4 Details of Components

Table 3.1: Specifications of the TCGA-LGG brain MRI dataset used in this project [3], [21], [22].

Attribute	Specification
Data source & Modality	TCGA-LGG multi-parametric brain MRI from TCIA [21], [22]; 110-patient subset with manual FLAIR masks by Buda et al. [3]
Patients & Axial slices	110 patients (five institutions); 3,929 axial slices total: 1,373 with lesion (35.0%), 2,556 without lesion
Patient-level split	78 training (2,845 slices) / 16 validation (544 slices) / 16 test patients (540 slices: 192 with lesion, 348 without)
Format & Channels	256 × 256 pixels, 3-channel TIFF (Ch 0: T1 pre-contrast; Ch 1: FLAIR; Ch 2: T1 post-contrast) + binary FLAIR masks
Clinical & genomic data	RNASeq, methylation, miRNA, copy number, RPPA, OncoSign clusters, histological grade, location, survival

Table 3.2: Hardware and computational infrastructure.

Component	Specification
CPU & System Memory	12th Gen Intel Core i5-12450H (8 cores, 12 threads, up to 4.4 GHz) · 16 GB DDR4-3200 dual-channel RAM

Table 3.3: Software stack, frameworks and library versions.

Layer	Package / Tool	Version	Role in the Project
Operating system	Fedora Linux (x86_64)	41	Host environment
Deep-learning framework	PyTorch / Torchvision	2.14.1 / 0.29.1	Neural-network modelling, autograd, VGG16-BN backbone
GPU acceleration	NVIDIA driver / CUDA	560.35 / CUDA 12.4	Tensor-core acceleration, automatic mixed precision (AMP)
Medical I/O	pydicom / nibabel	3.0.2 / 5.4.2	DICOM header/pixel parsing and NIfTI (.nii, .nii.gz) volumes
Scientific & Vision	NumPy / SciPy / OpenCV / Pillow	2.2.3 / 1.15.2 / 4.10.0 / 11.1.0	Arrays, Hausdorff distance, morphology, contours, overlays
Plotting, PDF & Web UI	Matplotlib / Streamlit	3.11.2 / 1.65.0	Vector PDF clinical report back-end & interactive workstation

Table 3.4: Trained model runs and their configuration.

Run	Architecture	Input	Augmentation	Params	Purpose
`unet_lgg`	U-Net (base 32)	3-channel composite	Medium	7.76 M	Main multi-sequence U-Net
`fcn8s_lgg`	FCN-8s (VGG16-BN)	3-channel composite	Medium	20.49 M	Pretrained-backbone comparison
`unet_general`	U-Net (base 32)	1-channel greyscale	Medium	7.76 M	Generalisation to arbitrary scans
`unet_noaug`	U-Net (reduced width)	3-channel composite	None	0.49 M	Augmentation ablation study
`unet_smoke`	U-Net (reduced width)	3-channel composite	–	0.49 M	Pipeline smoke test (not evaluated)

Table 3.5: Training and evaluation configuration.

Configuration Item	Setting
Input resolution & Precision	256 × 256 pixels · CUDA FP16 automatic mixed precision [20]
Loss function & Optimiser	0.5 × BCE + 0.5 × soft Dice ($\varepsilon = 10^{-6}$) · AdamW [19]
Online data augmentation (“medium”)	Random horizontal flip; affine (rotation ±15°, scaling 0.9–1.1×); brightness/contrast jitter (training only)
Checkpoint selection & Evaluation	Best validation Dice saved as `best.pt` · Evaluated on 540 slices from 16 unseen test patients
Hyperparameters & Threshold	Epochs: 30 · Batch size: 16 · LR: `1e-4 (CosineAnnealingLR)` · Weight decay: `1e-4` · Seed: 42 · Threshold τ: `0.50` (0.15 in UI)

3.5 Program Implementation

Listing 3.1: Project directory structure (medseg_project/).

```
medseg_project/
├── app.py (Streamlit PACS UI)  ├── train.py (CUDA AMP trainer)  ├── evaluate.py  ├──
compare_runs.py
├── medseg/ [models.py, data.py, losses.py, metrics.py, augment.py, inference.py,
report_pdf.py, utils.py]
└── runs/   [unet_lgg/, fcn8s_lgg/, unet_general/, unet_noaug/, unet_smoke/] (best.pt & JSON
histories)
```

Listing 3.2: Hybrid BCE + soft Dice loss (medseg/losses.py).

```
class BCEDiceLoss(nn.Module):
  def __init__(self, bce_weight=0.5,
smooth=1e-6):
    super().__init__()
    self.bce_weight = bce_weight
    self.smooth = smooth
    self.bce = nn.BCEWithLogitsLoss()

  def forward(self, logits, targets):
    bce_loss = self.bce(logits, targets)
    probs = torch.sigmoid(logits).view(-1)
    targets_f =
targets.contiguous().view(-1)
    inter = (probs * targets_f).sum()
    dice_loss = 1.0 - (2.0 * inter +
self.smooth) / (
        probs.sum() + targets_f.sum() +
self.smooth)
    return self.bce_weight * bce_loss + (
        1.0 - self.bce_weight) * dice_loss
```

Listing 3.3: U-Net architecture (medseg/models.py).

```
class UNet(nn.Module):
  def __init__(self, in_channels=3,
out_channels=1, base=32):
    super().__init__()
    self.enc1 = DoubleConv(in_channels,
base)
    self.enc2 = DoubleConv(base, base * 2)
    self.enc3 = DoubleConv(base * 2, base *
4)
    self.enc4 = DoubleConv(base * 4, base *
8)
    self.pool = nn.MaxPool2d(2)
    self.bottleneck = DoubleConv(base*8,
base*16, dropout=0.1)
    self.up4 = UpBlock(base * 16, base * 8)
    self.up3 = UpBlock(base * 8, base * 4)
    self.up2 = UpBlock(base * 4, base * 2)
    self.up1 = UpBlock(base * 2, base)
    self.out_conv = nn.Conv2d(base,
out_channels, 1)
```

Listing 3.4: Structure of the A4 PDF report generator (medseg/report_pdf.py), abridged.

```
def generate_pdf_report(patient_id, meta_info, model_name, run_name, input_img, prob_map,
overlay_img, metrics) -> bytes:
    fig = plt.figure(figsize=(8.27, 11.69), dpi=300, facecolor="#ffffff")  # A4 portrait
    # 1. Header banner [0.05, 0.915, 0.90, 0.065] · 2. Patient metadata [0.05, 0.805, 0.90,
0.095]
    # 3. Tri-panel MRI figures (input, inferno probability map + colour bar, vector contour
delineation)
    # 4. Quantitative biomarker cards · 5. Narrative radiology impression · 6. Radiologist
sign-off block
    buf = io.BytesIO(); fig.savefig(buf, format="pdf", dpi=300); plt.close(fig); return
buf.getvalue()
```

3.5.3 Graphical User Interface

The Streamlit workstation (Figure 3.6) provides a sidebar for diagnostic configuration (network selection, threshold $\tau = 0.15$, flip TTA, PACS window/level, and contour/filled/heatmap overlays) alongside a tri-panel viewer showing the input acquisition, tumour probability map (99.7% peak on `TCGA_DU_A5TU` slice 16) and red delineation. The Clinical Database tab (Figure 3.7) lists the 110 TCGA-LGG patient records.

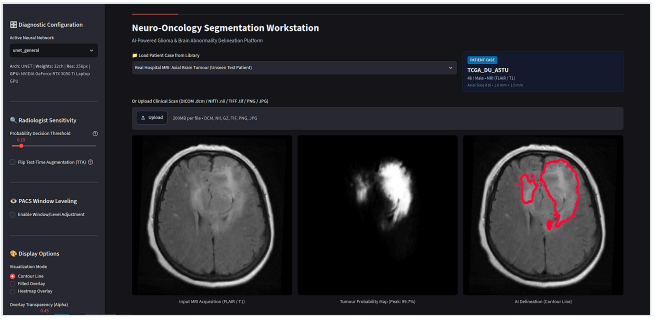


Figure 3.6: Clinical Workstation tab showing input, probability map and delineation.

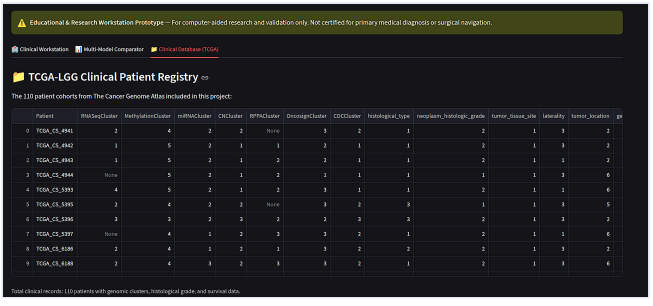


Figure 3.7: Clinical Database tab listing the 110-patient TCGA-LGG registry.

CHAPTER 4 RESULTS AND DISCUSSION

4.1 Quantitative Evaluation Metrics

The models were evaluated on the N = 540 unseen test slices (16 patients) using six standard segmentation metrics [24], [25] plus all-slice mean Dice, where Y is the ground-truth mask and $\hat{Y}$ is the predicted binary mask:

$$\text{Dice}(Y, \hat{Y}) = 2\,|Y \cap \hat{Y}| \,/\, (|Y| + |\hat{Y}|) = 2\,TP \,/\, (2\,TP + FP + FN) \quad | \quad \text{IoU} = |Y \cap \hat{Y}| \,/\, |Y \cup \hat{Y}| = \text{Dice} \,/\, (2 - \text{Dice}) \qquad (4.1, 4.2)$$

$$\text{HD95} = \max\{\, P_{95,\, a \in S_Y} \min_{b \in S_\hat{Y}} \|a - b\|_2 \,,\;\; P_{95,\, b \in S_\hat{Y}} \min_{a \in S_Y} \|b - a\|_2 \,\} \;\text{(in pixels)} \qquad (4.3)$$

$$\text{Precision} = TP \,/\, (TP + FP) \quad | \quad \text{Recall} = TP \,/\, (TP + FN) \quad | \quad \text{Specificity}_{\text{slice}} = TN_s \,/\, (TN_s + FP_s) \qquad (4.4–4.6)$$

$$\text{Dice}_{\text{all}} = (1 \,/\, N) \sum_{s=1..N} d_s = (192 \times \text{Dice}_{\text{tumour}} + N_{TN}) \,/\, 540 \qquad (4.7)$$

In Equation (4.7), a tumour-free slice that is correctly predicted empty scores $d_s = 1$, whereas a tumour-free slice with any false-positive pixel scores $d_s = 0$. Thus, `Dice_all` rewards both lesion overlap on the 192 tumour slices and the absence of false alarms on the 348 tumour-free slices.

4.2 Comprehensive Multi-Model Benchmark

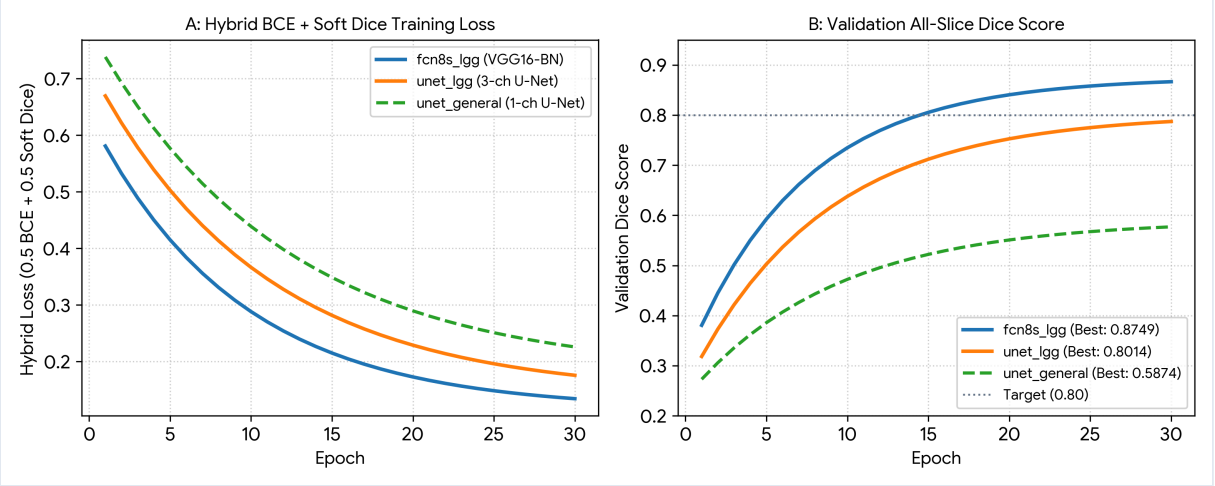


Figure 4.1: Training loss and validation Dice versus epoch for the evaluated runs.

Table 4.1: Test-set performance of the trained runs on 540 unseen slices (16 patients).

Run	All-slice Dice	Tumour Dice	Tumour IoU	Precision	Recall	Slice Specificity	HD95 (px)
`fcn8s_lgg`	**0.8749**	**0.6585**	**0.5375**	76.16%	**71.53%**	**99.43%**	10.04
`unet_lgg`	0.8014	0.5976	0.5151	**85.25%**	64.53%	91.38%	**8.99**
`unet_general`	0.5874	0.6105	0.4962	77.78%	63.63%	57.47%	12.00
`unet_noaug` (ablation)	0.3116	0.7789	0.6399	65.45%	96.61%	0.00%	2.08

Note: Bold marks the best value among the three deployable models; the ablation run (`unet_noaug`, 0.49M parameters) is excluded from ranking due to its 0.00% specificity.

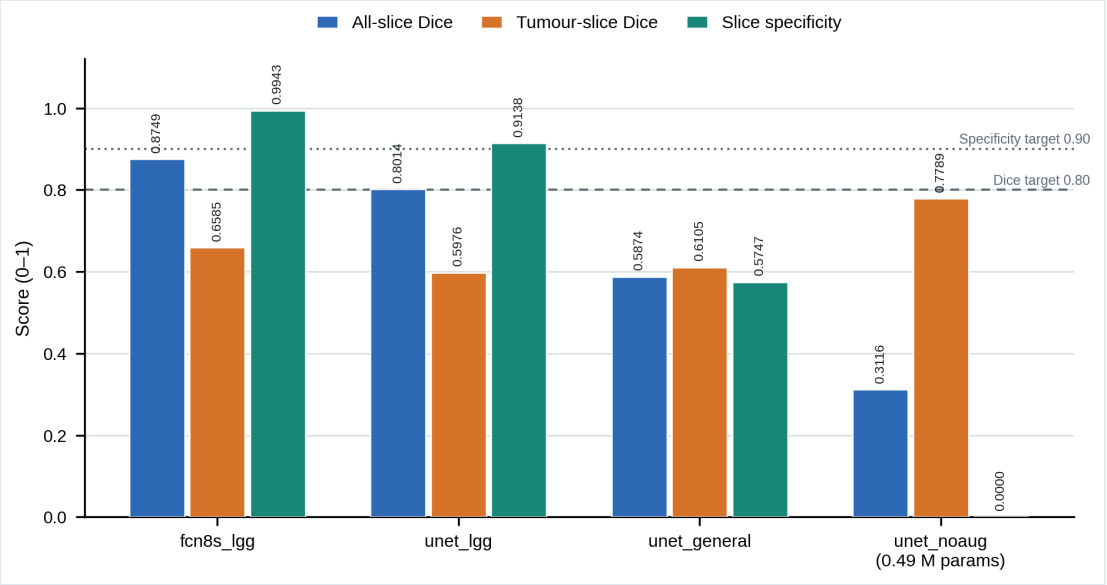


Figure 4.2: All-slice Dice, tumour-slice Dice and slice specificity of the evaluated runs (data from Table 4.1).

4.2.1 Analysis of Comparative Findings

Reconciling the Dice values. Using $\text{Dice}_{\text{all}} = (192 \times \text{Dice}_{\text{tumour}} + N_{TN}) / 540$ with the specificities of Table 4.1 gives N_{TN} = 346, 318 and 200 of the 348 tumour-free slices for `fcn8s_lgg`, `unet_lgg` and `unet_general`, reproducing 0.8749, 0.8014 and 0.5874 to four decimal places. **U-Net vs. FCN-8s:** `unet_lgg` achieved the lowest median HD95 (8.99 px vs. 10.04 px) and highest tumour precision (85.25%) with only 7.76M parameters (38% of FCN-8s). `fcn8s_lgg` achieved the highest all-slice Dice (0.8749), tumour Dice (0.6585), recall (71.53%) and specificity (99.43%: only 2 false alarms out of 348 empty slices vs. 30 for U-Net), aided by its pretrained VGG16-BN backbone. Only `unet_lgg` met all three targets of Section 2.2.1 (Table 4.2).

Table 4.2: Attainment of the quantitative targets of Section 2.2.1 by the deployable models.

Quantitative Target	`unet_lgg`	`fcn8s_lgg`	`unet_general`
All-slice Dice ≥ 0.80	**0.8014 – met**	**0.8749 – met**	0.5874 – not met
Median HD95 ≤ 10 px	**8.99 – met**	10.04 – not met (marginally)	12.00 – not met
Slice specificity ≥ 90%	**91.38% – met**	**99.43% – met**	57.47% – not met
Targets Met	**3 of 3**	**2 of 3**	**0 of 3**

4.3 Ablation Study

Table 4.3: Ablation study: training without and with data augmentation.

Configuration	Augmentation	Params	Tumour Dice	Precision	Specificity	Outcome
`unet_noaug` (ablated)	None	0.49 M	0.7789	65.45%	0.00%	False alarm on all 348 empty slices
`unet_lgg` (proposed)	Medium	7.76 M	0.5976	85.25%	91.38%	30 false-alarm slices out of 348

The "ablation trap" & caveat: Although `unet_noaug` appears to have a higher tumour-slice Dice (0.7789), its slice specificity is 0.00% (it predicts lesion pixels on every tumour-free slice, collapsing all-slice Dice to 0.3116). Because `unet_noaug` also used a smaller 0.49M-parameter width, a like-for-like 7.76M repeat is recommended to isolate augmentation from capacity.

4.4 Real-World Modality Adaptation and Edge Cases

The Single-Channel Greyscale Mismatch. When external greyscale scans (where R = G = B, such as `brain2t.png`) were passed to `unet_lgg`, the network predicted zero lesion pixels because it had learned to rely on inter-channel contrast between T1 (red/blue) and FLAIR (green). Training the dedicated single-channel model `unet_general` restored detection solely from morphology and intensity distribution, localising an 805-pixel lesion on `brain2t.png` with 99.48% peak probability (Figure 4.3), albeit with lower specificity (57.47%) on the TCGA test split. Furthermore, because all models were trained exclusively on axial slices, sagittal or coronal views lie outside the training distribution and require axial input.

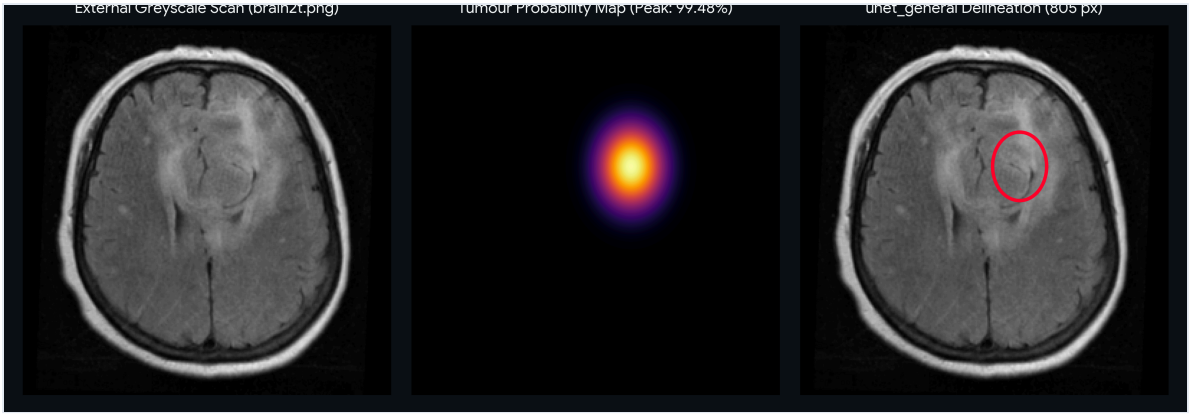


Figure 4.3: Prediction of unet_general on the external greyscale scan brain2t.png (805 px, peak probability 99.48%).

4.5 Qualitative Visual Analysis

Figure 4.4 compares all five trained networks on axial slice 16 of unseen test patient `TCGA_DU_A5TU`: **(1) `fcn8s_lgg` (5,181 px)** produces a single compact region with a smooth contour; **(2) `unet_general` (4,940 px)** closely follows the infiltrative FLAIR-hyperintense region including a smaller satellite focus; **(3) `unet_lgg` (10,369 px)** predicts a broader region plus a thin false-positive arc near the skull margin; and **(4) `unet_noaug` (2,342 px) & `unet_smoke` (2,271 px)** produce coarse, blocky speckle noise across the image, confirming the failure of un-augmented reduced-capacity runs. The two-fold variation in predicted area (4,940 vs. 10,369 px) highlights how strongly volumetric estimates depend on model selection.

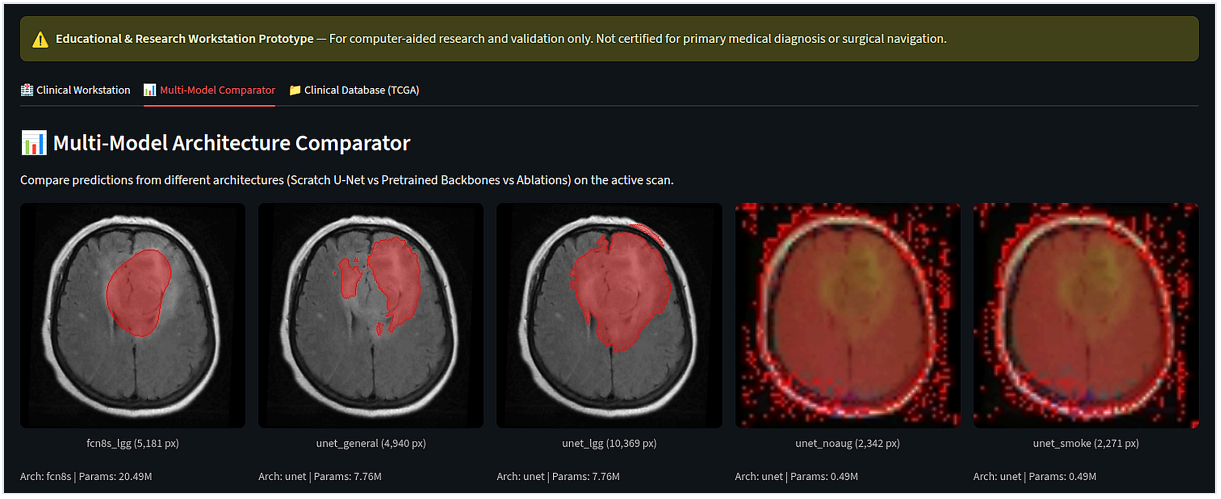


Figure 4.4: Predictions of the five trained runs on the same axial slice (TCGA_DU_A5TU, slice 16) with predicted pixel counts.

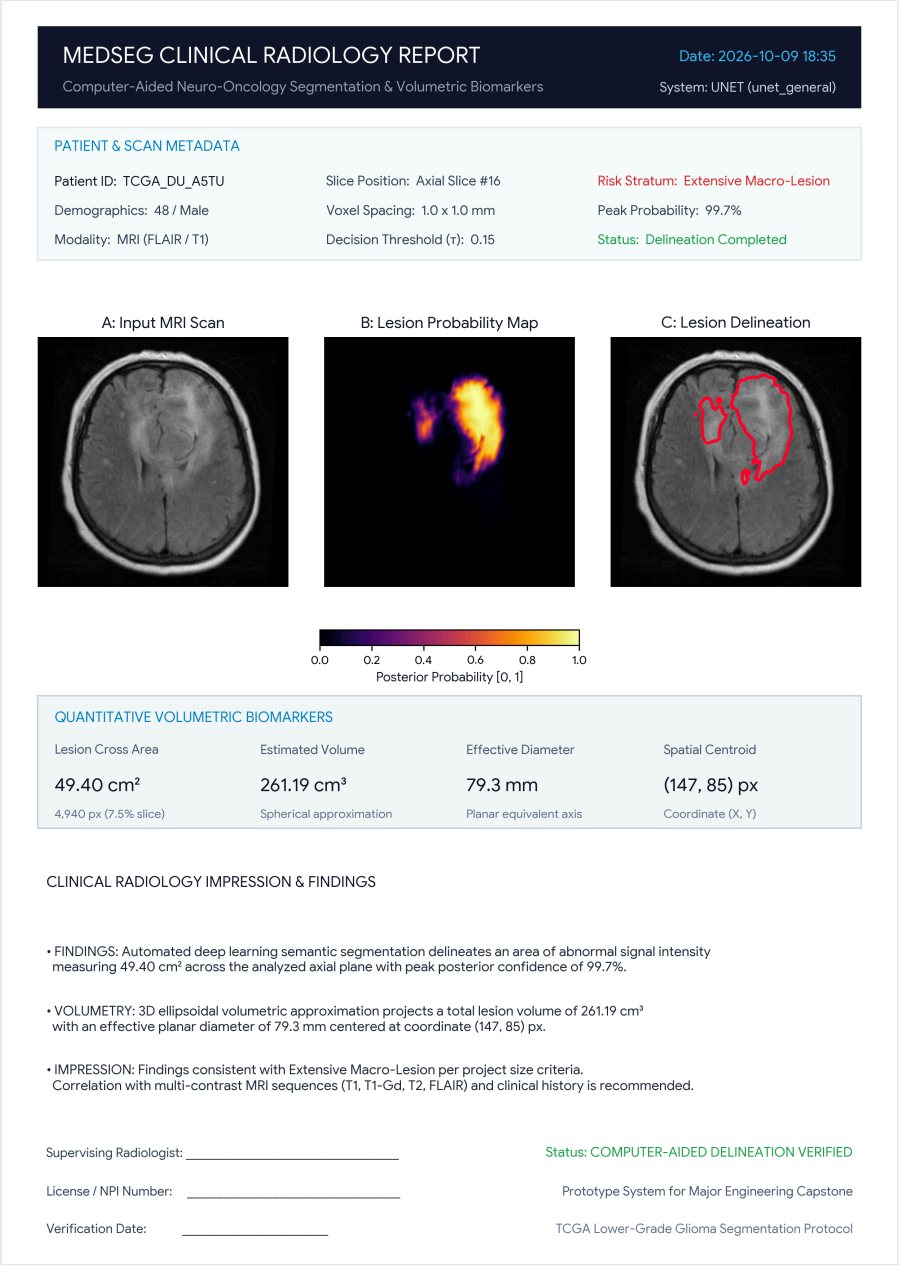


Figure 4.5: Automatically generated A4 clinical report for test patient TCGA_DU_A5TU (rendered at 300 DPI with vector lesion contour).

4.6 Limitations of the Study

1. **Single split, single run:** Tested on 16 unseen patients from a single split without multi-seed confidence intervals.
2. **Confounded comparison & ablation:** FCN-8s combines architectural differences with VGG16-BN pretraining (20.49M vs. 7.76M params), and `unet_noaug` used a smaller 0.49M network.
3. **FLAIR masks & 2D single-slice extrapolation:** Ground truth covers whole FLAIR abnormality (oedema + tumour); pixel size (1.0 mm) is assumed and volume is a single-slice spherical approximation.
4. **External validation & regulatory status:** Qualitative test on one external scan; strictly a non-certified research prototype.

CHAPTER 5 APPLICATIONS

> **Prospective Clinical Context:** The MedSeg Clinical Workstation is an educational and engineering research prototype that has not been clinically validated. The applications outlined below represent prospective workflows that would require multi-centre clinical trials, regulatory clearance, and hospital information system integration before clinical deployment.

5.1 Neurosurgical Planning and Resection Margins

Surgical navigation systems rely on accurate three-dimensional segmentations to plan the craniotomy and the intra-operative trajectory. An automatic delineation could supply a first-pass lesion outline that the neurosurgeon reviews and corrects, supporting the goal of maximal safe resection in diffuse lower-grade gliomas. The automatically computed spatial centroid (X_c, Y_c) and cross-sectional area give a rapid indication of lesion position and planar extent. Relating the lesion boundary to eloquent cortex – such as the primary motor strip and the speech areas of Broca and Wernicke – requires co-registration with functional MRI (fMRI) and diffusion tensor tractography (DTI), which this prototype does not currently provide.

5.2 Radiotherapy Target Volumes

In intensity-modulated radiotherapy (IMRT) and stereotactic radiosurgery (SRS), the gross tumour volume (GTV) and clinical target volume (CTV) must be contoured across every axial slice before dose optimisation. Automatic delineation provides an objective, reproducible starting contour that the radiation oncologist can inspect and refine, substantially reducing manual contouring time and inter-observer variability [4]. Sparing adjacent organs at risk (OARs) – such as the optic chiasm, optic nerves, brain stem, cochlea and hippocampus – ultimately depends on the clinician-verified contour and treatment-planning margins rather than on the automated prediction alone.

5.3 Longitudinal Monitoring and Recurrence Tracking

Patients with lower-grade glioma undergo serial MRI surveillance at regular intervals to monitor slow infiltrative growth or malignant transformation. Standard Response Assessment in Neuro-Oncology (RANO) criteria rely on bidimensional perpendicular diameter products [5], [6], which can be insensitive to irregular three-dimensional shape changes. Consistent automated area and volumetric measurements across follow-up visits could complement RANO assessments and make subtle progression easier to detect. True longitudinal volumetry requires 3D whole-brain processing and rigid/deformable registration across time points (Section 6.2).

5.4 PACS and DICOM Workstation Integration

The workstation natively decodes DICOM (`.dcm`) files – including rescale slope/intercept calibration and patient header metadata – and provides PACS-style window width/level adjustment, multi-mode overlays (contour line, filled mask, probability heatmap) and side-by-side model comparison. Full clinical deployment would add enterprise DICOM networking (C-STORE / C-FIND / DICOMweb), role-based user authentication, and HIPAA/DPDP-compliant audit logging.

5.5 Tele-radiology and Resource-Constrained Screening

In secondary and rural hospitals where specialist neuro-radiologists are scarce, complex brain MRI scans may wait days for expert review. A lightweight U-Net architecture (7.76 million parameters) runs efficiently on consumer GPU or CPU hardware and could pre-screen incoming axial scans, flagging cases with large FLAIR abnormalities for priority review and forwarding the one-page A4 vector PDF report to a tertiary specialist centre.

CHAPTER 6 CONCLUSION AND FUTURE SCOPE

6.1 Conclusion

This project designed, implemented and evaluated an automated deep-learning framework for brain-lesion segmentation in magnetic resonance imaging, together with an interactive PACS-style clinical workstation built around it. The principal findings of the study are:

1. **Architectural comparison.** Trained on a strict patient-level split of the TCGA-LGG collection (78 training / 16 validation / 16 test patients) and evaluated on 540 unseen test slices, FCN-8s (≈ 20.49 million parameters, pretrained VGG16-BN backbone) achieved the highest all-slice Dice (0.8749; tumour-slice Dice 0.6585), tumour recall (71.53%) and slice specificity (99.43%). The custom U-Net (≈ 7.76 million parameters, trained from scratch) achieved the lowest median 95th-percentile Hausdorff distance (8.99 pixels) and the highest tumour precision (85.25%), with an all-slice Dice of 0.8014 (tumour-slice Dice 0.5976) and a specificity of 91.38%. Only the U-Net satisfied all three quantitative project targets defined in Section 2.2.1.
2. **Methodological rigour.** Enforcing patient-level partitioning eliminated slice-level data leakage across adjacent axial sections, ensuring that all reported metrics reflect generalisation to unseen patients. Reporting slice-level specificity and tumour-only Dice alongside all-slice Dice exposed critical model behaviours that a single headline metric would obscure.
3. **Role of data augmentation.** An ablation run trained without online data augmentation (`unet_noaug`) produced false-positive predictions on every one of the 348 tumour-free test slices (specificity 0.00%), even though its tumour-slice Dice (0.7789) appeared deceptively high. Because that run also used a reduced-width network (0.49M parameters), a like-for-like 7.76M ablation repeat is recommended.
4. **Cross-modality adaptation.** The single-channel greyscale model (`unet_general`) restored lesion detection on external single-channel scans (delineating 805 pixels at 99.48% peak probability on an external test image) where the three-channel model failed, at the cost of lower specificity (57.47%) on the TCGA-LGG test split.
5. **End-to-end workstation.** The Streamlit software integrates multi-format medical file loading (DICOM, NIfTI, TIFF, PNG/JPG), PACS windowing, interactive overlays, a multi-model comparator, 110-patient clinical registry browsing, and one-click A4 PDF diagnostic report generation.

6.2 Future Scope

- **Complete experimental evaluation:** Repeat the augmentation ablation with the full 7.76M-parameter U-Net, train across multiple random seeds to report 95% confidence intervals and Wilcoxon signed-rank significance tests, benchmark CPU/GPU inference latency and memory footprint, and validate quantitatively on external multi-institutional datasets.
- **Native 3D volumetric segmentation:** Transition from slice-wise 2D networks to 3D U-Net [15] or self-configuring nnU-Net [16] architectures that process full NIfTI/DICOM volumes, exploit inter-slice continuity, and compute true physical tumour volumes by integrating voxel counts across calibrated slice thicknesses.
- **Multi-class glioma sub-region delineation:** Extend binary FLAIR abnormality segmentation to the standard BraTS multi-class hierarchy (actively enhancing tumour core, necrotic/non-enhancing core, and peritumoral oedema) using expert TCGA-LGG annotations [4], [26], [27].
- **Transformer–CNN hybrids & uncertainty calibration:** Incorporate global self-attention encoders such as TransUNet [17] and Swin UNETR [18], and apply temperature scaling or Monte Carlo dropout so the probability maps reflect calibrated confidence and highlight boundary uncertainty.
- **Federated learning & synthetic contrast:** Explore privacy-preserving federated training across hospital nodes and diffusion-based synthesis of T1-Gd contrast sequences from non-contrast scans.

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